



Department of MCA
CIE-I Second Semester

Subject Name: Agile Software Engineering

Subject Code: 25 MCA220

Time: 1 hr

Max. Marks: 20

Course Outcomes:

CO1:	Explain various software characteristics and analyze different software Development Models
CO2:	Demonstrate the contents of a SRS and apply basic software quality assurance practices to ensure that design, development meet or exceed applicable standards
CO3:	Compare and contrast various methods for software design
CO4:	Formulate testing strategy for software systems, employ techniques such as unit testing, Test driven development and functional testing
CO5:	Manage software development process independently as well as in teams and make use of Various software management tools for development, maintenance and analysis.

PART A: ANSWER ALL THE QUESTIONS Each question carries 1mark 4Qx1=04

Q	QUESTIONS	BL	CO
1	a Define Software Engineering.	1	CO1
	b A software project is delayed due to poor planning and increasing costs. Which concept describes this situation?	2	CO1
	c What is Software Requirement Specification (SRS)?	1	CO2
	d Explain Verification in Software Quality Assurance?	2	CO2

PART B: ANSWER ALL THE QUESTIONS Each question carries 4 marks 3QX4=12

Q	QUESTIONS	BL	CO
2	a Illustrate the phases of the Waterfall Model by applying them to a simple software development project.	3	CO1
	OR b A bank wants to develop a new online banking interface but the users are not sure about their exact requirements. Explain how the Prototype Model can be used in this situation and describe the steps involved?	3	CO1
3	a Apply the Iterative Enhancement Model to a real-world software project and explain how the system is developed through multiple iterations and improvements.	3	CO1
	OR b Examine the phases of the Spiral Model and analyze its advantages compared with the Waterfall Model.	4	CO1
4	a Design an Entity Relationship (ER) diagram for a Library Management System showing entities, attributes, and relationships?	3	CO2
	OR b A company is developing a Payroll Management System that processes employee details, calculates salaries, and generates reports. Identify the main processes, inputs, outputs, and data stores and explain how a Data Flow Diagram (DFD) level 1 can represent this system.	3	CO2



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PART C: ANSWER ANY 1 QUESTION

Each question carries 4 marks

1Qx4=4

Q	QUESTION	BL	CO
1	Analyze the phases of the Software Development Life Cycle (SDLC) and explain how each phase contributes to the development of a high-quality software system	4	CO1
2	Apply the concepts of Software Quality Assurance (SQA) to explain how verification, validation and SQA plans ensure software reliability and performance.	3	CO2



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Department: Computer Applications
MCA- Second Semester
CIE-2

Subject Name: Agile Software Development
Time: 1 hrs.

Subject Code: 25 MCA220
Max. Marks: 20

Course Outcomes:

- CO1:** Explain various software characteristics and analyze different software Development Models
CO2: Demonstrate the contents of a SRS and apply basic software quality assurance practices to ensure that design, development meet or exceed applicable standards
CO3: Compare and contrast various methods for software design.
CO4: Formulate testing strategy for software systems, employ techniques such as unit testing, Test driven development and functional testing
CO5: Manage software development process independently as well as in teams and make use of Various software management tools for development, maintenance and analysis.

PART A: ANSWER ANY ALL THE QUESTIONS Each question carries 1 marks **4Qx1=04**

Q	QUESTIONS	BL	CO
1	a A system has high number of inputs, outputs, and files. Which size-oriented metric is suitable?	1	3
	b A program has many decision points leading to multiple paths. Which metric helps measure complexity?	2	3
	c Software is tested by developers before integration. Identify the testing level. <i>A</i>	1	4
	d Testing is done after combining modules, and errors occur due to interaction. Identify the testing level. <i>T</i>	2	4

PART B : ANSWER ALL THE QUESTIONS Each question carries 4 marks **3QX4=12**

Q	QUESTIONS	BL	CO
2	a Write the difference between top down and bottom up design? Give an example.	4	3
	OR		
	b Apply Function-Oriented and Object-Oriented design approaches in software development. Illustrate how each approach is used in real system design.	3	3
3	a Explain control flow graph? And calculate the cyclomatic complexity using the formula and shown the flow graph? Display_Value(int a , int b) { While(a>b) { If(a>0) a=a-b; else b=b-a; } Print a , b; }	4	3
	OR		
	b Consider a software project with the following information domain characteristic for the calculation of function point metric. Number of external inputs (I) = 9 Number of external output (O) = 6	4	3



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		Number of external inquiries (E) = 3 Number of files (F) = 8 Number of external interfaces (N) = 70 It is given that the complexity weighting factors for I, O, E, F, and N are High, respectively. It is also given that, out of fourteen value adjustment factors that influence the development effort, four factors has value 4, each of the other four factors has value 5, and each of the remaining factors has value 11. The computed value of the function point metric is _____. a) Compute Unadjusted Function Points (UFP) b) Calculate Value Adjustment Factor (VAF) c) Find final FP value d) Explain why two systems with same UFP can have different FP		
4	a	Apply White Box Testing, Black Box Testing, Alpha Testing, and Beta Testing in a software project and explain how each technique is used at different stages of testing to improve overall software quality.	3	4
	OR			
	b	Explain pseudo- codes and Write " Binary search " Pseudo code:	3	4

PART C : ANSWER ANY 1 QUESTIONS Each question carries 4 marks

1Qx4=4

Q	QUESTIONS	BL	CO
5	Analyze the software design process and evaluate how design decisions impact software quality. A) Compare architectural design and low-level design in terms of flexibility, maintainability, and scalability. Also analyze the role of modularization and its effect on coupling and cohesion.	4	3
6	Analyze the different levels of testing (Unit, Integration, Acceptance, Regression) and evaluate how each level contributes to overall software quality. Identify potential risks if any one level is skipped.	4	4